

Differentiating Vectors

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The change in a vector $\vec{A}$ over a time interval Δt is:

$$\Delta \vec{A} = \vec{A}(t + \Delta t) - \vec{A}(t)$$

The derivative of a vector is defined as:

$$\frac{d\vec{A}}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{A}}{\Delta t}$$

Two Special Cases

Case 1: $\Delta \vec{A}$ is parallel to $\vec{A}$

When the change in $\vec{A}$ is parallel to itself, the magnitude changes but the direction remains constant:

$$\begin{aligned} \lim_{\Delta t \rightarrow 0} \frac{|\vec{A}(t + \Delta t)| - |\vec{A}(t)|}{\Delta t} &\neq 0 \\ \lim_{\Delta t \rightarrow 0} \frac{|\hat{A}(t + \Delta t)| - |\hat{A}(t)|}{\Delta t} &= 0 \end{aligned}$$

Case 2: $\Delta \vec{A}$ is orthogonal to $\vec{A}$

Consider $\Delta \vec{A} = \Delta t \beta \hat{k}$ where $\Delta \vec{A} \perp \vec{A}$.

The change of magnitude at $t = 0$ is:

$$\begin{aligned} \frac{d|\vec{A}(t)|}{dt} &= \lim_{\Delta t \rightarrow 0} \frac{|\vec{A}(0) + \Delta \vec{A}| - |\vec{A}(0)|}{\Delta t} \\ &= \lim_{\Delta t \rightarrow 0} \frac{\sqrt{|\vec{A}(0)|^2 + \beta^2(\Delta t)^2} - |\vec{A}(0)|}{\Delta t} \\ &= \left. \frac{d}{dt} \sqrt{|\vec{A}|^2 + \beta^2 t^2} \right|_{t=0} \\ &= \left. \frac{\beta^2 t}{\sqrt{|\vec{A}|^2 + \beta^2 t^2}} \right|_{t=0} \\ &= 0 \end{aligned}$$

Conclusion: For small Δt , the magnitude $|\vec{A}(t + \Delta t)|$ does not change if $\Delta \vec{A} \perp \vec{A}$.

Magnitude of the Derivative (Orthogonal Case)

When $\Delta \vec{A} \perp \vec{A}$, we can express the magnitude of the derivative geometrically:

$$\left| \frac{d\vec{A}(t)}{dt} \right| = \lim_{\Delta t \rightarrow 0} \frac{|\vec{A}| \tan(\Delta \theta)}{\Delta t}$$

To evaluate this limit:

$$\begin{aligned}
\lim_{\Delta t \rightarrow 0} \frac{\tan(\Delta\theta)}{\Delta t} &= \lim_{\Delta t \rightarrow 0} \frac{\tan(0 + \Delta\theta) - \tan(0)}{\Delta\theta} \cdot \frac{\Delta\theta}{\Delta t} \\
&= \left. \frac{d}{d\theta} [\tan(\theta)] \right|_{\theta=0} \cdot \frac{d\theta}{dt} \\
&= \left. \sec^2(\theta) \right|_{\theta=0} \cdot \frac{d\theta}{dt} \\
&= \frac{d\theta}{dt}
\end{aligned}$$

Thus, we obtain the important result:

$$\left| \frac{d\vec{A}(t)}{dt} \right| = |\vec{A}| \frac{d\theta}{dt}$$

This shows that when $\vec{A}$ changes only in direction (not magnitude), the rate of change depends on the angular velocity $\frac{d\theta}{dt}$.

Combining the two cases

For vectors whose rates of change are neither orthogonal nor parallel to the vector, we can decompose the changes into orthogonal and parallel components:

$$\left| \frac{d\vec{A}_\perp}{dt} \right| = |\vec{A}| \frac{d\theta}{dt} \quad (\text{changes direction})$$

$$\left| \frac{d\vec{A}_\parallel}{dt} \right| = \frac{d|\vec{A}|}{dt} \quad (\text{changes magnitude})$$

where the perpendicular component controls direction change and the parallel component controls magnitude change.